

PAPER ONLY

Alpaca MCP Server v2

Alpaca AI Trading Agents 2026

EVIDENCE-GATED AUTONOMOUS TRADING

EdgeStack

Every rule survived an attempt to **kill it**.

An agent whose strategy was built backward: months spent trying to disprove candidate edges — and it only trades what survived.

33 years

OF VALIDATION DATA

3

BACKTEST ENGINES FORCED TO
AGREE

110

RESEARCH SCRIPTS, ALL PUBLIC

1

GRAVEYARD OF REJECTED IDEAS

LLM trading agents lose money **confidently** — their rules come from vibes.

Ask a model for a strategy and it gives you one: plausible, articulate, untested. The field's fix is to wrap the model in deterministic guardrails — and this event is full of excellent guardrails. But guardrails only decide *whether a proposal is allowed*. Nobody asks the harder question:

What evidence does a trading rule need before it deserves to exist?

The graveyard is the point.

- **Elliott waves** — shuffled-return surrogates reproduce the "patterns" (corr -0.525 vs -0.526)
- **Fibonacci retracements** — 38.2/50/61.8 rank 4th–14th of 28 arbitrary bands
- **Intraday mean reversion** — the "edge" was bid-ask bounce (trade AC -0.43 , midquote $+0.07$)
- **Leveraged-ETF decay shorting** — the drag is real ($t = -5.4$) and drift swamps it 10:1
- **Five macro overlays** — four contradicted out-of-sample, including one we had already explained
- **Our own first options design** — negative expectancy from quoting the worst fill. Found, measured, fixed.

Every rejection is documented with the number that killed it. The negative results are load-bearing: they are why the survivors can be trusted.

Three components. Every number is a measurement.

COMPONENT	RULE	EVIDENCE
Overnight core	Long SPY close→open only, gated by 12-month trend AND the credit canary	Sharpe 0.89 overnight vs 0.05 intraday · 8/9 eras since 1993
Capitulation sleeve	Buy 5σ panics on heavy-but-not-extreme volume (1.4–2.5×), 7 ETFs, 3-session hold	+1.42%/event · 67.6% win · t=4.27 · 136 events / 33y · surrogate-tested
Options component	Defined-risk bull put spreads behind 14 deterministic gates	trades the same validated signal; sizing capped by measured friction — bounded by evidence, not vibes

The index earns its return while the market is closed.



Sharpe ratio, 1993–2026. $\text{corr}(\text{overnight}, \text{intraday}) = 0.004$ — dropping the intraday half is risk reduction, not relabeling.

$t(\text{diff})=6.58$, out-of-sample, $n=4,359$. The same boundary appears in the volume ceiling and the credit canary: **markets revert emotional moves and honor informational ones.**

Every parameter tuning **failed validation**. The untuned rules won — and that is the point.

CONFIG	TRAIN SHARPE (2008–17)	VALIDATION SHARPE (2018–26)	VALIDATION DD
always-on core (no gates)	0.51 · DD –50.7%	—	—
research defaults	0.80	0.65	21.2%
"improved" (tuned on train)	0.89	0.44	28.1%
+ credit canary (from the trader's own older strategy — adopted only after passing BOTH windows)	0.98	1.02	11.8%

A rule change counts only if it survives a disjoint window. One did. Full protocol: ENGINE-TRIAL.md.

Signals are arithmetic. Gates are pure functions. The LLM narrates — it never sizes, prices, or authorizes.

- **Production engine reproduces the 33-year research record to 3 decimals** — research and live code cannot drift apart silently (test_signal_engine.py)
- **14 deterministic risk gates**, 22-case suite where the *right* gate must object — a gate that never fires is not a control
- **Alpaca MCP Server v2 integration is real**: account reads and order submission route through MCP (streamable-http), REST fallback logged per call
- **Append-only decision journal** — every session recorded, including refusals, with the evidence for each
- **Self-hosted, unattended**: ensure-running supervisors, logon persistence, public live dashboard

EdgeStack • LIVE

Every rule survived an attempt to kill it · 33 years of evidence, three backtest engines, one graveyard · Alpaca paper account PA3ZCDD0PR2N

EQUITY

\$100,000

+0 vs \$100k start

EQUITY GATE

CLOSED

trend UP (12m +19.0%) | credit
DETERIORATING (HYG 79.74 vs SMA100
79.85)

MACRO REGIME

TLT 21d sd 0.555 vs 90d mean 0.760
(ratio 0.73) -> CALM

ALPACA MCP SERVER

ROUTING

orders + account via MCP, REST
fallback

OPEN POSITIONS

SYMBOL	COMPONENT	SIZE	ENTERED
flat — waiting for signals that clear the gates			

STRATEGY — EVERY NUMBER IS A MEASUREMENT

- SPY overnight core · Sharpe 0.89 vs 0.05 intraday · 8/9 eras
- gate: 12-month trend AND credit canary (HYG > SMA100) · validated on disjoint windows 0.80→0.98 / 0.65→1.02
- capitulation sleeve · +1.42%/event · 67.6% win · t=4.27 · 33y
- volume ceiling 2.5x — above it "real news arrived", edge dies
- options: defined-risk put spreads behind 14 deterministic gates

DECISION JOURNAL (LATEST SESSIONS)

SESSION	SIGNALS	WHAT HAPPENED & WHY
2026-08-30	0	core_gated: trend UP (12m +19.0%) credit DETERIORATING (HYG 79.74 vs SMA100 79.8
2026-08-30	0	no trade — gates held
2026-08-30	0	no trade — gates held
2026-08-30	0	no trade — gates held

The equity gate is **CLOSED**: trend is up, but credit slipped below its 100-day average — risk is being repriced on information, so the agent refuses. **Refusal is a decision with evidence attached.**

A real user, an honest model, a pre-registered path to real capital.

- **User** — the retail systematic trader who wants agent autonomy with institutional discipline (the segment Alpaca's API business already monetizes)
- **Model** — strategy-as-infrastructure on brokerage rails; the evidence pipeline, not another subscription chatbot
- **Next edge** — earnings implied-move premium (measured $\sim 1.33\times$ realized, $t=3.95$) enters the same kill-test pipeline
- **Real capital gate** — pre-registered: 60 live sessions, realized Sharpe within 1σ of backtest, zero gate violations
- **Team** — built solo, in seven days, with an AI pair that had to show its work

Rules that **earned their existence.** An engine that **provably implements them.** An agent that **explains every refusal.**

What we won't pretend: these are risk-adjusted edges, a 5-session P&L window is mostly noise, and flat is a position. An entry that hides its limits is asking judges to find them — we measured ours first.

jpennin5.github.io/edgestack · github.com/jpennin5/edgestack · Alpaca paper PA3ZCDD0PR2N

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